

Aritro Saha

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Experience

Software Engineer 1

Jul 2025 – Present

Bristol Myers Squibb

Hyderabad, India

- Reduced diagnostic review time for NSCLC cases by **30%**, by architecting a medical diagnostic assistant with LLM-driven clinical decision support.
- Reduced release cycle time by **40+** engineering hours per cycle, by architecting a company-wide Databricks accelerator with standardized ETL/ELT microservices and YAML-driven CI/CD pipeline generation
- Owned end-to-end delivery of a self-service LLM marketplace plugin adopted by **20+** internal teams, sharply reducing recurring support tickets by replacing manual deployment requests with automated workflows
- Scaled data ingestion to handle **5M+** records per hour by building serverless, distributed AWS Lambda pipelines with Glue-crawler metadata enrichment and DynamoDB persistence
- **Technologies:** Python, Spark Databricks, SQL, AWS (Lambda, S3, DynamoDB, Glue, Bedrock, SageMaker, OpenSearch, Redshift)

Data Science Engineer Intern

Feb 2025 – Jun 2025

Bajaj Finserv Health

Pune, India

- Cut OPD claims processing failure rate by **60%**, by redesigning a distributed, vision-based NER pipeline scaled to **40,000+** claims/day
- Designed and implemented a regex-based NER mapping service for ICD-10 codes, eliminating third-party API dependencies and accelerating throughput by **98%**
- Improved P99 backend latency by **35% (800ms to 520ms)**, by refactoring FastAPI endpoint serialization and adding unit test coverage across critical paths
- **Technologies:** Python, FastAPI, Elasticsearch, MongoDB, Google Gemini, OpenAI, Azure DevOps

Software Engineering Intern

Oct 2023 – Dec 2023

WIPRO

Kolkata, India

- Built a 3D visualization engine in Three.js with optimized asset loading, making client-side render times roughly **3x** faster
- Increased deployment frequency by **50%**, by automating CI/CD pipelines with Docker and Azure DevOps
- **Technologies:** React.js, Node.js, Three.js, Git, Docker, Azure

Projects

Structurify: ETL Data Pipeline | GCP, LangGraph, Gemini, Firebase, Terraform

[GitHub](#) | [Live](#)

- Engineered a serverless fan-out architecture on GCP utilizing Cloud Run and Pub/Sub, decoupling API gateways to process datasets of up to **1M rows**, eliminating server timeouts and ensuring **99%** availability under high-concurrency burst loads
- Architected a fault-tolerant Map-Reduce ETL pipeline using LangGraph and Gemini AI with dynamic batching, utilizing a retry state machine to correct LLM hallucinations and achieve **99.8%** schema validation accuracy
- Integrated DuckDB for high-speed distributed data aggregation, accelerating the compilation of multi-vendor chunks by over **40%** while logging atomic LLM token usage in Firestore for accurate enterprise billing
- Designed a real-time Next.js client synchronized with Firestore listeners, delivering **sub 200ms** live processing updates and secure artifacts for multi-tenant workspaces authenticated via Firebase Identity Platform

Quarantine | Python, AWS, Azure, GCP, CI/CD, Conda

[GitHub](#) | [PyPI](#)

- Engineered a resilient error-handling library for Python ETL pipelines, implementing a dynamic dead-letter queue that prevents batch failures and ensures continuous execution for loops of **10,000+** items
- Architected a fault-tolerant processing engine with AWS, Azure and GCP integration to isolate malformed records, reducing debugging and data recovery time for large-scale workflows by over **40%**
- Automated multi-platform distribution via GitHub Actions CI/CD pipelines, deploying the package to both **PyPI** and **Conda-Forge** while maintaining strict semantic versioning and greater than **90%** test coverage

Education

Vellore Institute of Technology

Sep 2021 – Jul 2025

Bachelor of Technology in Electronics and Computer Engineering | **CGPA: 8.53/10.0**

Chennai, India

Publications & Achievements

Competitive Programming | LeetCode

Peak Rating: 1868

- Solving **630+** questions, achieved Knight Badge status, placing in the top six percentile of active competitive programmers globally.

KrishnaVision: A Multimodal Virtual Interface for Context-Aware HCI

[\[View Paper\]](#)

International Journal of Innovative Research in Technology (IJIRT)

- Introduced a virtual mouse system achieving 97.3% gesture recognition accuracy at 22ms latency while reducing power usage by 41% compared to ResNet-50 baselines.